PLEASE READ THIS COVER PAGE AND FOLLOW THE INSTRUCTIONS.
DO NOT OPEN THE EXAM UNTIL YOU ARE TOLD TO DO SO.

On the computer graded answer sheet in the appropriate boxes, enter your:

1. **name**
2. **student identification number**
3. **Lab Section** (In the four columns labeled "Section Id", using a zero before your recitation section number. Ex., section 231 is written as 0231.)

Then **fill in the corresponding bubbles** below your name, ID number, and recitation section.

Answer all questions on the computer graded answer sheets by filling in the proper bubble with a No. 2 pencil. If you change an answer, erase the undesired mark thoroughly. Mark only the best answer to each question. Extra time will not be permitted at the end of the exam to fill in your scantron bubble sheet so bubble in your answers as you go.

Cell phones must be turned off; all electronic devices must be placed in a backpack or on the floor and may not be accessed during the exam. Students violating this policy will be asked to leave and will receive a zero for the exam.

A Periodic Table is printed on the back of this cover sheet. There are 6 exam pages (double-sided) with 25 questions. Each question is worth 4 points for a total of 100 points. There are 2 additional bonus questions worth 2 points each. Answer these as questions 26 and 27. Scratch paper is attached to the back of the exam. When you are instructed to begin the exam, please check that you have all the pages.

When you are finished, give the answer sheet to your teaching assistant or the proctor in the room. The scantron is your official record of answers. You may take the exam with you. Answers will be posted on Canvas at the completion of the exam. Good luck!

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

Potentially Useful Information:

$$Density = \frac{mass}{volume}$$

$$K = 273.15 + \ ℃$$

$$℃ = \frac{5}{9}(°F - 32)$$

$$°F = \left(\frac{9}{5}℃\right) + 32$$

$$E = h\nu$$

$$c = \lambda\nu$$

$c = 2.998 \times 10^{8}$ m/s

$$F = K_e \frac{q_1 q_2}{r^2}$$

$$Z_{eff} = Z - shielding\ electrons$$

$h = 6.626 \times 10^{-34}$ J*s

1 amu = 1.66×10^{-24} g

1 km = 0.612137 miles

1 kg = 2.204 lbs

1 L = 0.264 gal

0.3047 m = 1 ft

1 lb = 453.592 g

$1cm^3$ = 1cc = 1 mL

N_A =Avogadro's Number = 6.022×10^{23}

$$Molarity = \frac{moles\ solute}{Liters\ of\ Solution}$$

$M_1V_1 = M_2V_2$

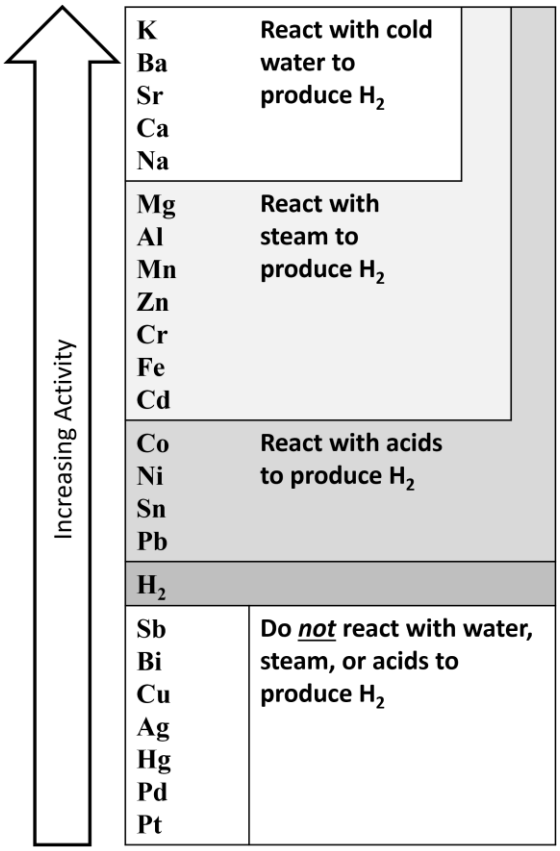
The <u>7 strong acids</u> are: HCl, HBr, HI, H_2SO_4, HNO_3, $HClO_4$, and $HClO_3$

Strong Bases: All Soluble hydroxide salts

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

Activity Series:



Some Common Polyatomic Ions

Ammonium	NH_4^+
Hydronium	H_3O^+
Hydroxide	OH^-
Cyanide	CN^-
Carbonate	CO_3^{2-}
Phosphate	PO_4^{3-}
Chlorate	ClO_3^-
Bromate	BrO_3^-
Iodate	IO_3^-
Sulfate	SO_4^{2-}
Selenate	SeO_4^{2-}
Nitrate	NO_3^-

Electronegativity Table

H 2.1																	He ---
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne ---
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.2	S 2.5	Cl 3.0	Ar ---
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr 3.0
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe 2.6
Cs 0.7	Ba 0.9	La-Lu 1.1-1.2	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn ---
Fr 0.7	Ra 0.9	Ac-No 1.1-1.7															

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

Solubility Rules

Ion	Rule	Exceptions
Na^+, K^+, NH_4^+	Most salts containing an alkali metal or ammonium ion are soluble	None
NO_3^-, $CH_3CO_2^-$	All nitrates and acetate are soluble	None
SO_4^{2-}	Most sulfates are soluble	$BaSO_4$, $SrSO_4$ $PbSO_4$ $CaSO_4$, Hg_2SO_4 and Ag_2SO_4
Cl^-, Br^-, I^-	Most chlorides, bromides and iodides are soluble	AgX, Hg_2X_2, PbX_2, and HgI_2 (X= Cl, Br, or I)
Ag^+	Silver compounds are generally insoluble	$AgNO_3$ and $AgClO_4$ are soluble $AgCH_3CO_2$ is slightly soluble
O^{2-}, OH^-	Oxides and hydroxides are insoluble	Alkali metal hydroxides, $Ba(OH)_2$, $Sr(OH)_2$ $Ca(OH)_2$ is somewhat soluble
S^{2-}	Sulfides are insoluble	Exceptions to this rule are compounds with Na^+, K^+, NH_4^+ and alkaline earth metal cations.
CrO_4^{2-}	Most chromates are insoluble	Compounds containing Na^+, K^+, NH_4^+, Mg^{2+}, Ca^{2+}, Al^{3+}, and Ni^{2+}
CO_3^{2-} , PO_4^{3-}, SO_3^{2-}, SiO_3^{2-}	Most carbonates, phosphates, sulfites, and silicates are insoluble	Compounds containing Na^+, K^+, NH_4^+

Potentially Less Useful Information:

Strawberries actually contain less sugar than lemons (when measured in percent mass)! The citric acid molecules overwhelm taste buds and hide how much sugar is present.

Almonds are a member of the rose family which also includes peaches and raspberries.

Johnny Appleseed's apples weren't actually intended to be eaten. They were much more bitter, and their purpose was to be made into hard apple cider.

The Moon has moonquakes! They are suspected to be caused by tidal stresses from the Earth.

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

The Periodic Table of Elements

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.012											5 B 10.81	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.085	15 P 30.974	16 S 32.06	17 Cl 35.453	18 Ar 39.948
19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.38	31 Ga 69.723	32 Ge 72.63	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.798
37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.96	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57-71 Lanthanides	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89-103 Actinides	104 Rf (267)	105 Db (268)	106 Sg (269)	107 Bh (270)	108 Hs (269)	109 Mt (278)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (294)	118 Og (294)

57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97
89 Ac (227)	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

1. **Returning Topic:** Report the answer to the following calculation with the correct number of significant figures:

$$\frac{6.0509 * 5.000}{9.8 + 4.70}$$

A. 2
B. 2.1
C. 2.09
D. 2.086

2. **Returning Topic:** How many square centimeters are there in 3.19 square meters?

A. 3.19×10^{-4} square centimeters
B. 3.19×10^{-2} square centimeters
C. 3.19×10^{2} square centimeters
D. 3.19×10^{4} square centimeters

3. **Returning Topic:** Consider the transition metal cobalt (Z=27). How many d electrons are in the ground state electron figuration of the Co^{2+} ion?

A. 9
B. 2
C. 7
D. 5

4. **Returning Topic:** Which of the following is the correct electron configuration for a neutral molybdenum (Z=42) atom?

A. $[Kr]5s^2 4d^4$
B. $[Kr]5s^2 4d^5$
C. $[Kr]5d^5$
D. $[Kr]5s^1 4d^5$

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

5. Manganese forms the following compound with sulfur, MnS_3. What is the name of this compound?

   A. manganese sulfide
   B. manganese trisulfide
   C. manganese(I) trisulfide
   D. manganese (VI) sulfide

6. Which of the following ions would form the strongest bond with a fluoride ion?

   A. Cs+
   B. K^+
   C. Ba^{2+}
   D. Ca^{2+}

7. Consider the compound S_4Cl_2. What is the correct name for this compound?

   A. sulfur chloride
   B. tetrasulfur bichloride
   C. tetrasulfur dichloride
   D. quadrasulfur dichloride

8. The element ralphium reacts with chipine to form a compound with the name ralphium pentachipide. Which of the following statements about this compound is ***correct***?

   A. Ralphium is a non-metal element.
   B. Ralphium is further right or above chipine on the periodic table.
   C. The compound ralphium pentachipide contains three times as many chipine atoms as ralphium atoms.
   D. chipine is a metal.

9. Categorize the following compounds as ionic or covalent.

   I: NaBr II: N_2O III: NH_4I

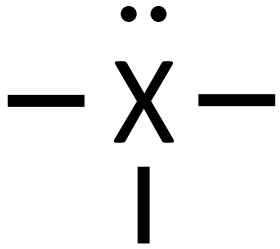
   A. I is ionic; II and III are covalent.
   B. I and III are ionic; II is covalent.
   C. I, II, and III are all covalent.
   D. I and III are covalent; II is ionic.

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

10. Which of the following bonds is most polar?

    A. O-O
    B. N-O
    C. Cl-O
    D. C-O

11. What is the electron domain geometry on the atom X in the structure shown to the right.

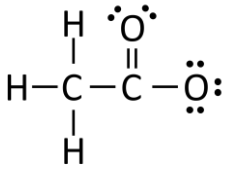


    A. Trigonal Planar
    B. Tetrahedral
    C. Linear
    D. Trigonal Pyramidal

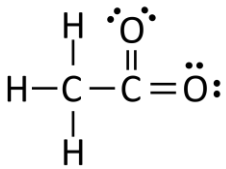
12. Predict the molecular shape and the approximate bond angle of oxygen difluoride.

    A. Bent, 109.5º
    B. Trigonal planar, 120º
    C. Tetrahedral, 109.5º
    D. Bent, < 109.5º

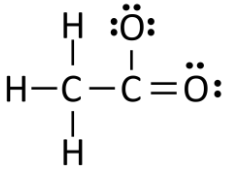
13. The acetate ion, $CH_3CO_2^-$, exhibits resonance. The first resonance structure is shown below. Which structure is another resonance structure for the acetate ion?



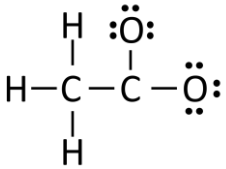
A.



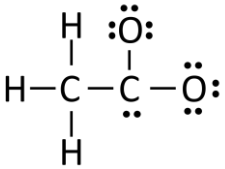
B.



C.



D.



This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

14. You prepare an aqueous solution of sodium bromide. Which of the following statements about this solution is correct?

    A. The solvent is water, and the solute is NaBr.
    B. The solvent is NaBr and the solute is water.
    C. The solvent is water and the solute is SBr_2.
    D. The solvent is SBr_2 and the solute is water.

15. You need to prepare 160.0 mL of a 1.85 M solution of NaCl. How many moles of NaCl do you need?

    A. 0.296 moles of NaCl
    B. 11.6 moles of NaCl
    C. 296 moles of NaCl
    D. 0.0116 moles of NaCl

16. Use the solubility rules to determine which of the following compounds are ***not*** soluble in water.

    A. $(NH_4)_2S$
    B. $MgSO_4$
    C. CaS
    D. $PbCl_2$

17. Adamantine (Ad) is a fictional non-metal that forms an acid with the formula HAd. What is the name of this compound?

    A. Hydroadamantic Acid
    B. Hydroadamanous Acid
    C. Adamantic Acid
    D. Adamantous Acid

18. Nervine (Nr) is a fictional element that forms the polyatomic ion named nervate with the formula NrO_3^{2-}. What would be the name of the acid with the chemical formula H_2NrO_4?

    A. Pernervous Acid
    B. Pernervic Acid
    C. Hyponervic Acid
    D. Hyponervous Acid

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

19. Which of the following soluble compounds when dissolved in water will ***not*** create an electrolyte solution?
    A. HBr
    B. CH_4
    C. $Ba(OH)_2$
    D. Na_2O

20. Consider the following chemical reaction:

$$CuCO_3\,(s) \xrightarrow{heat} CuO\,(s) + CO_2\,(g)$$

Which of the following statements about this reaction is ***incorrect***?

    A. $CuCO_3$ is a reactant.
    B. CuO and CO_2 are products.
    C. All of the reactants and products are in the same phase when this reaction to occurs.
    D. This reaction requires heat to proceed.

21. Balance the following chemical equation:

$$S_8\,(s) + O_2\,(g) \rightarrow SO_3\,(g) \qquad (unbalanced)$$

How many ***atoms*** of oxygen are found on the ***product side*** when the equation is properly balanced?

    A. 3
    B. 8
    C. 12
    D. 24

22. Phosphorus tribromide reacts with elemental fluorine gas to produce tetraphosphorus pentafluoride and elemental bromine gas. Write a properly balanced chemical equation for this reaction. How many ***fluorine molecules*** are consumed in this reaction?

    A. 5
    B. 8
    C. 12
    D. 1

This content is protected and may not be shared, uploaded, or distributed to any 3rd party without express written permission of the course instructor.

23. Consider the following chemical reaction:

$$4\ P + 3\ Cl_2 \rightarrow 2\ P_2Cl_3$$

This reaction would be classified as a ____.

A. Single Displacement Reaction
B. Double Displacement Reaction
C. Synthesis (Combination) Reaction
D. Decomposition reaction

24. Which of the following chemical reactions will ***not*** occur?

A. $2\ K\ (s) + FeCl_2\ (aq) \rightarrow 2\ KCl\ (aq) + Fe\ (s)$
B. $Ni\ (s) + 2\ NaCl\ (aq) \rightarrow NiCl_2\ (aq) + 2\ Na\ (s)$
C. $3\ Sn\ (s) + 2\ CuCl_3\ (aq) \rightarrow 3\ SnCl_2\ (aq) + 2\ Cu\ (s)$
D. $Mg\ (s) + 2\ AgNO_3\ (aq) \rightarrow Mg(NO_3)_2\ (aq) + 2\ Ag\ (s)$

25. Which of the is not a ***product*** of the complete combustion of a hydrocarbon?

A. CO_2
B. H_2O
C. Heat
D. O_2